

ROAD ACCIDENT PRONE ZONES IDENTIFICATION SYSTEM

Mehak Garg
Department of Computer Science
Chitkara University Institute of
Engineering and Technology
Rajpura, Punjab, India
mehak572.be22@chitkara.edu.in

Monika Aggarwal
Department of Computer Science
Chitkara University Institute of
Engineering and Technology
Rajpura, Punjab, India
monika.aggarwal@chitkara.edu.in

ABSTRACT – Road accidents cause problems as they are responsible for death rates, injuries and economic damage. Many researches are already done to study accident patterns by collecting data but these studies did not help us and were unable to provide information when needed. Here we have used GIS mapping along with machine learning to predict accidents by looking at multiple factors such as location, time of accident, weather conditions and injuries caused. We have used K-Means clustering and DBSCAN density based clustering to find areas where accident rate is high. The system is built using MERN stack along with Python – MongoDB as database, Express.js & Node.js as backend technologies and React.js as frontend part. This model gives 80-81% accuracy approximately which is far better than the old models. These technologies help in making the system user friendly which makes it easier for users to use and give them safety tips. This system is very useful for real life and help people to be careful on road. The system try to reduce accident rates by providing necessary information.

KEYWORDS – Road Accident Prone Zone Detection, GIS Mapping, MERN stack, Python, Clustering Algorithms, Classification Techniques, Web Visualization, Road Safety.

1. LITERATURE REVIEW

Road accidents rate is increasing rapidly causing rise in death rates, injuries and economic damage. Earlier researchers are done on the basis of numerical and statistical data to check for patters in accidents. These methods ware good at finding patterns but they lacked to get information on how different things worked together [4]. With advancement in technology, GIS based systems were used to look at locations on map where accidents happened. Researchers like Chen said that GIS is essential for looking at data and marking it on maps [1]. But even with these maps we could not predict when accident would happen in future. Next people started using machine learning to find accident prone zones. They started using classification algorithms including Logistic Regression, Decision Tree and Random Forest to predict when accidents might happen by considering multiple factors.

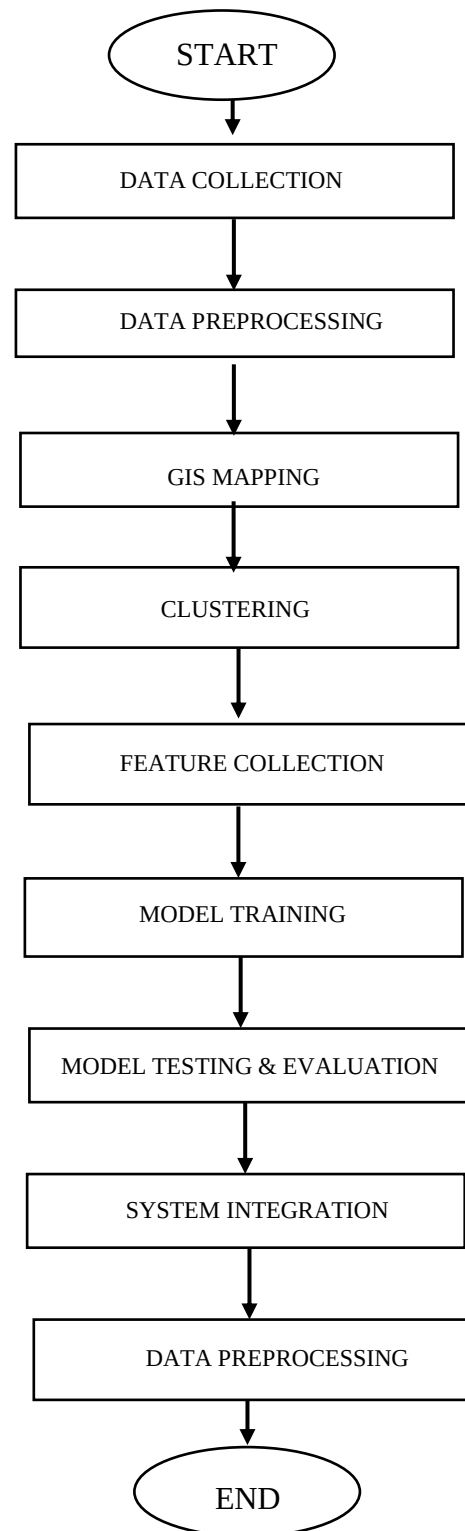
Kumar and others found that machine learning models help in improving prediction results more accurately than the old methods [2]. Gupta and Verma showed that Random Forest provide better performance than other methods and is better at predicting how bad accidents are [9]. Random Forest provide 80% accuracy approximately depending on how good the data is. They have also used clustering techniques such as K-Means and DBSCAN

to find dense regions where accident rate is high. Li and his team helped in explaining how K-Means groups accident prone areas into clusters based on how similar they are while Wang showed that DBSCAN is better for this data because it helps in finding dense areas and handling noise well [3][8]. Factors like the weather and road conditions also affect when accidents happen and how bad they are. After all these improvements most of the systems only look at wither predicting or marking locations on map not both [6]. Zhang said that combining GIS and machine learning together could help in making analysis better [5]. This paper is about building a system that brings clustering, machine learning and interactive mapping together all in a single platform. This system also focuses on making the solution user-friendly and interactive.

2. **OBJECTIVES**

This paper is about building a tool that helps in finding areas where accident rate is very high on the basis of historical data. It also uses machine learning model to know how much risky it could be to drive through such areas. The goal is to have everything in one place for analyzing and predicting so that it would be easier to use and more practical than the old methods. Another important thing is build a user-friendly website so that users can see accident prone zones and make better choices about the travel which will help in reducing accident rates. We also want make the prediction results more correct by considering multiple factors that could affect accidents such as weather, location, road conditions, time, road traffic and accident severity. The system is designed such that future enhancements could be made like adding real time data, smart alerts and advanced tools to make analyzing better to keep model updated and help people to understand actual accident rates so that they can make better choices accordingly.

3. **METHODOLOGY**



This flowchart help us to understand how the system actually works from start to end. The process starts from collecting accident data and ends with user interaction. The

accident data is collected from a trusted source Kaggle under Indian road accident dataset or any other public platform could be used to collect dataset. This dataset contains information about accidents like location, times, weather conditions, accident severity and causes. After collecting the data, it is cleaned by removing any missing information, duplicate records and things that do not make any sense so that the dataset becomes more accurate for analysis. Once it is done, then we use GIS mapping to mark accident locations on map using their coordinates which contains their latitude and longitude. After marking we have used clustering algorithms such as K-Means and DBSCAN density based algorithm to group accident points together. K-Means helps in grouping points that are close to each other in form of clusters. DBSCAN density based algorithm helps in finding areas with a high number of accidents and ignores any noise. This way we can see where a lot of accidents happen and where accidents are spread out in areas. After clustering we choose the important factors that affect accidents such as weather, time and severity. This step helps in making the model better by focusing only on important things. On finishing training the model, we test it against data to see how the model actually works. This is evaluated using performance metrics. This helps us know that the model not only performs well on existing data but it also works well on new data. Once all these steps are done, we put together all the parts in one system during integration phase by connecting frontend, backend and machine learning together. As a result, a system is available for users to see accident areas on a map and analyse how risky that area could be. The process ends by providing a system that is practical, easy to use and performs well in the real world. We have used past accident data to train the model accordingly and at the end it helped us understand the data more clearly. By following a particular structure, the end result is created for users to use in practical situations.

4. RESULTS AND ANALYSIS

The result and analysis part helps us to show that the accident prone zone identification system can detect areas with high accident rates and mark their locations on map using their coordinates. The system uses clustering algorithms that work well in highlighting places where accident occurs frequently making it easier to identify high risk zones. This helps users to see which areas requires and allows them to relate the accident data to real locations. The system is using Random Forest which provides accuracy of 80% approximately that is far better compared to old methods and simpler machine learning models. The system results in good accuracy because it look for factors like location, weather conditions, time and accident severity. The system is able to uncover more meaningful patterns in the data due to combination of analysis along with predictive machine learning model. Another important result is that the system can find high risk zones which makes the system more dependable and useful in situations. Users can depend on the result completely and take action accordingly.

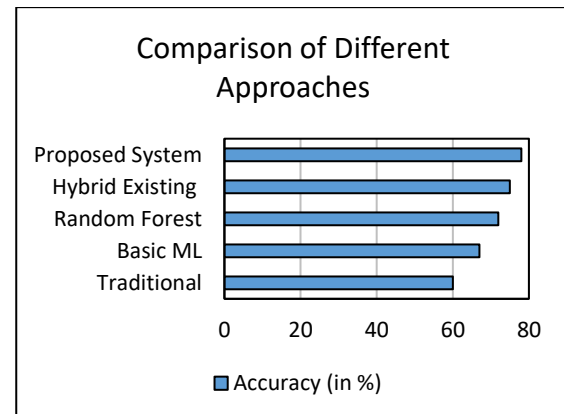


Fig.1. Comparison of Different Approaches

Fig.1. displays the comparison of how well the different approaches analyse and predict road accidents based on their accuracy percentage. Traditional methods gives an accuracy of 60% which is not good as they depend only on statistical techniques and cannot handle complex relationship in accident data. Basic machine learning

model are slightly better than traditional methods. These models learn patterns from data but still not great at considering multiple factors that affect accidents. The Random Forest model performs better with an accuracy of around 72%. It combines decision trees and handles how different features interact more efficiently. The Random Forest model uses decision trees. Hybrid system that combines machine learning with some integration, like GIS have an accuracy of around 75%. However they often do not fully integrate all components. Hybrid systems use machine learning and GIS. The proposed system has the accuracy of around 78%. It uses clustering algorithms like K-Means and DBSCAN machine learning, like Random and GIS-based visualization. This combination helps the system identify areas to accidents and predict risk levels more effectively. The proposed system uses clustering algorithms and machine learning.

PERFORMANCE METRICS		
S.NO	METRICS	VALUE (in %)
1.	Accuracy	78
2.	Precision	76
3.	Recall	75
4.	F1 Score	76

Table 1: Performance results from evaluated machine learning model

Table1. evaluates the effectiveness of the system on the basis of accuracy, precision, recall and F1 score. These metrics provide a comprehensive understanding of the model's prediction.

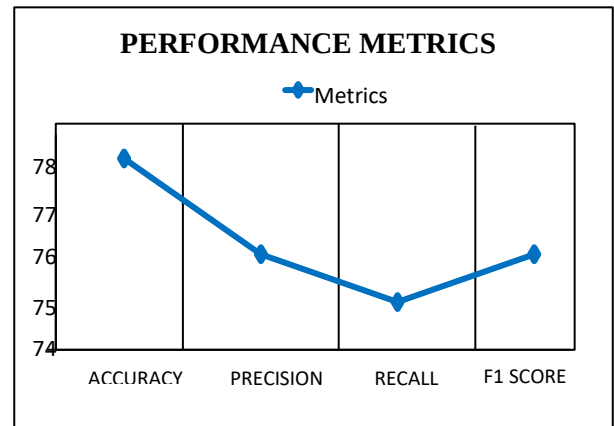


Fig.2. Performance Metrics Chart

Fig.2. evaluates the effectiveness of the system on the basis of accuracy, precision, recall and F1 score. These metrics provide a comprehensive understanding of the model's prediction. Accuracy is defined as the proportion of predicted observations to the total number of observations and the system achieves accuracy of approximately 78% which results in better performance. Precision measures how many of the predicted high risk zones are actually correct. The system shows a precision of 76% which indicates that the predicted high risk zones are accurate. Recall measures how well the model identifies actual high-risk zones. With a recall of 75%, the accident prone identification system successfully detects the majority of real accident prone areas minimizing missed risks. The F1 score shows how well precision and recall are balanced. It tells us that the model does a job of finding high risk zones without making too many mistakes. The values shows that the system is balanced and dependable for real situations. The system makes predictive analysis which results in making the system more practical and effective for accessing accident risk. Using clustering and machine learning together provides better insights than applying the methods individually. Clustering helps in identifying patterns and regions with large number of accidents while the prediction model adds more accuracy by estimating risk levels. They create a more complete and meaningful understanding of accident data.

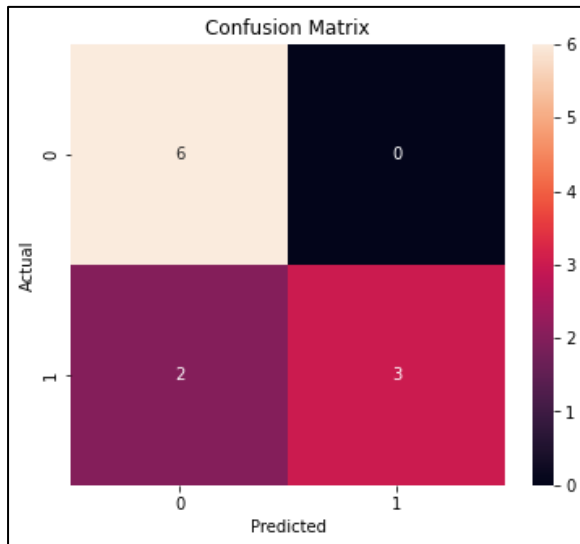


Fig.3: Confusion Matrix for road accident prone zones identification system model

Fig3. evaluates model performance by looking at the accident risk classifications that actually happened and the ones the model predicted. In this study the model was tested against 11 samples. Out of these, 6 instances were correctly identifies as low risk areas which shows the model is really good and accurate at recognizing areas that are safe. 3 instances were correctly identifies as high risk areas which means the model can detect areas where accidents are more likely to happen. The model did not make any mistakes by saying an area was risk when it was actually safe. This is a thing because it does not send out unnecessary warnings. However the model did make a mistake by saying 2 areas were low risk when they were actually high risk. This means the model missed some areas that are not safe. The results show that the model is very good at not making mistakes when it says an areas is a low risk. The models performance metrics show this too where the model is very precise. But not as good at finding all the high risk areas. It can be improved so it does not miss as many high risk areas as the accident risk model does. Overall the integration of these techniques improves both the ability to analyse and the power to make predictions of the system. The system not only performs well in terms of accuracy but also

gives users a simple and accurate way to understand the outcomes. All these results make the solution more effective, practical and suitable for actual usage.

5. IMPLEMENTATION DETAILS

The system is made using a stack approach so that all parts work smoothly together and give users a smooth experience. The frontend is made with React.js, which helps make an interactive interface where users can easily see and understand accident data. The interface is so easy to use so people who don't know much about tech can use the system without any problems. The backend is made with Node.js and Express.js. It is used to handle tasks and manage API requests. This helps to transfer data between the frontend and the database. The backend handles user requests. It sends required data to the frontend. It connects the machine learning model to the web application. The server takes care of what people put into the website sends the data back to them and connects the accident prediction part to the website. It also makes sure multiple users can use the system at the time without affecting performance. This makes the website reliable and fast. We use MongoDB to store information about accidents. It is good at handling a large amount of data. It can store many different kinds of information like where accidents happen what the weather is like what time it happens and how bad the accident is. The database is setup to make it easy to add information and get the information we need quickly so the website can handle more data as we get it. The accident prediction part is made using Python. We train a model called Random Forest on some data that we have cleaned up. Once training is completed the model is build as a single system. It provides predictions when user uses the platform. For map visualization tools various available APIs are used. These tools display accident locations on a map. High risk areas are highlighted using colors. Users can work

with the map by making bigger or smaller and picking places on it. The map system is really useful and simple to use. Markers and pop-ups are used to show accident related information. The system focuses on performance, accuracy and easy to use which makes the system really good for things that happen in life. The system is made up of separate parts which makes it easy to add new features to the system.

6. CONCLUSION

The system displays how GIS, clustering and machine learning all together can be used to analyse and predict accidents. We have combined visualization and prediction in a single platform which results in offering a more complete solution than old methods that only look at historical data. The system is not only good at predicting accidents but also make the results easy to understand as it provides an interactive interface. Users can look for accident patterns and high risk areas without the need of technical knowledge. The system considers factors like location, weather, time and accident severity helping in producing more realistic predictions. Clustering technique is also used to find areas that are more frequent to experience accidents. The system is easy and practical to use. The simple interface and map helps users to look at data and get information quickly saves time and effort. Modern technologies makes it easy to solve real problems. By combining various approaches in a single platform the system provides a complete solution. Further improvements can be made by using real data to give predictions up to date. Features like alerts, route suggestions and integrating it as a mobile app. Overall this system is a step towards making roads safe and reducing accident rates if some improvements are made.

7. REFERENCES

1. CHEN ET AL. (2019) said that we use GIS to look at accident data on maps. It makes it easy to look at patterns to find accidents areas.
2. KUMAR ET AL. (2021) looked at machine learning to predict road accidents. The models can do a job of predicting accidents old methods of doing things.
3. LI ET AL. (2020) studied how to group data. They showed how methods such as K-Means can help find accident hotspots by looking at data points.
4. ANDERSON AND JOHNSON (2018) thought it was really important to know what happened with accidents. They found out that looking at the past is very important to understand accidents.
5. ZHANG ET AL. (2022) looked at how combining GIS and machine learning can make both looking at and predicting accidents which makes analyzing accidents more effective.
6. SINGH AND SHARMA (2020) looked at how things like the weather and the condition of the roads can affect how bad an accident is.
7. PATEL ET AL. (2021) made a system on the internet to track accidents. They said it is very important that this system is easy for people to use so that many people can use it. This way people can easily report accidents and use the system without any trouble.
8. WANG ET AL. (2019) used the DBSCAN clustering algorithm to find areas where a lot of accident happen. They showed it works with real data.it consider
9. GUPTA AND VERMA (2022) used the Random Forest algorithm. They helped to predict how severe an accident would be. They found that combining methods can give accurate result.
10. World Health Organization (2023) provided us with information about road safety. They said technology is needed to prevent accidents.

11. PEDREGOSA ET AL. (2011) introduced the Scikit-learn which is used in old models that work in real life applications.

12. Kaggle Dataset for Indian Road Accident

<https://www.kaggle.com/datasets/parthdevrani/india-road-accident>

13. OpenStreetMap Contributors (2024) gave us source data. This information helps us in making maps and is important for building GIS applications.